

BALASUBRAMANIAN SHANMUGHAM

SENIOR BACKEND ENGINEER, JAVA | SPRING BOOT

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SUMMARY

15 years of backend engineering experience building enterprise applications at scale in Java, Spring Boot using microservice patterns including database-per-service, design circuit breaker and using asynchronous messaging. Deep experience in JVM and storage fine-tuning, designing event-driven, high-throughput systems with strong observability and distributed tracing.

EXPERIENCE

Senior Backend Engineer — Java | Bahwan CyberTek Pvt Ltd | Jul 2018 – Present

Continuous employment; the entries below reflect concurrent client engagements as a services-model senior backend engineer, not sequential roles.

- Designed and deployed containerised microservices on Kubernetes via Red Hat OpenShift, running multiple pods per service for horizontal scalability and fault tolerance
- Built event-driven, loosely-coupled services using Kafka for asynchronous communication between microservices
- Applied cloud-native resilience patterns — circuit breakers, retries, bulkheads — to prevent cascading failures across distributed services
- Used Docker for consistent, reproducible deployments across environments
- Integrated CI/CD pipelines (GitHub Actions) for automated build, test, and deployment workflows
- Deployed and ran production workloads on AWS, with Grafana and CloudWatch for observability, alerting, and automated recovery
- Applied Redis-based distributed caching and clustering to support high-throughput, low-latency service performance
- Designed secure, multi-tenant REST APIs with OAuth2/JWT and RBAC, deployed as independently scalable services
- Diagnosed and resolved production performance issues using JMeter for load testing and JVM profiling tools to isolate bottlenecks such as thread contention, slow queries, and memory pressure
- Worked 8+ years in fully remote, cross-geography teams, using Jira/Linear for sprint and task tracking, Slack for real-time and async communication, and structured documentation practices to stay productive and accountable across time zones

PROJECTS

Centralised Job Scheduler | Bahwan CyberTek Pvt Ltd | Feb 2025 – Present

- Designed and implemented multiple scheduler job types (fire-and-forget, recurring, and chained jobs)
- Applied pessimistic DB-level locking on job state-transition actions (pause, resume, stop, delete)
- Implemented circuit-breaker patterns to prevent cascading failures and error propagation to downstream systems
- Reduced P99 latency by tuning cache eviction TTLs and cutting lock contention on job state-transition actions
- Introduced Kafka-based asynchronous fallback processing with batch processing, increasing the throughput
- Managed CI/CD workflows via GitHub Actions with a peer code review process enforced on every change

RRTS / NCRTC — Passenger Information Display System | Bahwan CyberTek Pvt Ltd | Jan 2022 – Dec 2024

- Designed the passenger flow for a train ticketing platform (search → seat selection → payment confirmation → invoice/receipt → booking notifications) serving ~1M users/day
- Implemented a background timeout/expiry job to automatically release provisionally held seats if payment wasn't completed in time, preventing inventory lock-up
- Migrated the booking flow to an event-driven Saga pattern using Kafka, with a dedicated consumer releasing held seats on payment-failure events
- Configured a dead-letter queue with 7-day retention and 2 retries on exponential backoff, with reprocessing logic for train-availability, payment-processing, and seat-allocation failures based on the booking's last completed step
- Designed Java socket-based real-time communication for live train movement and schedule data (Operational challenge)
- Built a modular, containerised Spring Boot application template for ticketing, passenger info, and route management

ETL Pipeline — Fintech (Tranzact) | Bahwan CyberTek Pvt Ltd *Jul 2018 – Dec 2022*

- Designed Apache Spark ELT pipelines processing streaming Kafka data into a Hadoop-based lakehouse (HBase and Hive), monitored using the Ambari framework
- Worked with RDD and DataFrame transformations, applying partitioning strategies to optimise distributed processing across the Spark cluster
- Tuned Spark executor and worker-thread allocation, alongside serialisation and batch settings, to improve throughput on high-volume workloads
- Built a reusable Scala tool to bypass underperforming Spark RDD transformations and directly ingest records from HBase to Hive during manual reprocessing
- Optimised PySpark and Spark SQL query execution for large-scale data processing
- Integrated Grafana and AWS CloudWatch for pipeline observability and automated recovery

EARLIER CAREER

Senior Java Developer | Genpact Pvt Ltd | *May 2014 – Jun 2018*

- Built a high-throughput CRM interaction-ingestion engine and reporting system (Email Auto-logging)
- Used Java concurrency and database partitioning to improve batch-write performance by 4x

Contract Placement — IBM Java Technology Centre | Synova Pvt Ltd | *Feb 2010 – May 2012*

- Worked on security API integration for IBM's Tivoli product line, focused on PKCS#11 and JCA/JCE cryptographic API implementation on z/OS using JZOS

Contract Placement — Detica | Digital Harbor Pvt Ltd | *May 2012 – Dec 2013*

- Feature and bug fix implementations on existing Watchlist and Scenario Manager modules for a credit card anti-money-laundering platform, integrating OFAC and Factiva for sanctions and adverse-media screening

SKILLS & CERTIFICATIONS

Technical Skills: Java, Spring Boot, Spring Cloud (AWS, GCP), Node.js, multi-threading & concurrency, JVM and DB tuning, Kafka, Redis, Docker, Kubernetes, OAuth2/JWT, SQL (Postgres) & NoSQL (MongoDB, DynamoDB)

Certifications: Advanced Cloud Computing, Blockchain & IoT — IIT Chennai

Awards: WeMakeDevs Hackathons • AMD Developer Program • MLH Member

INDEPENDENT PROJECTS

portfolio.balashan.dev

getConsenso — <https://www.contractkit.balashan.dev> (will be moved to www.getconsenso.com)

Contract-first API governance platform (OpenAPI/GitHub scanning, self-healing contract validation, approval workflow, multi-language stub generation). Built with Next.js; uses Java/Mustache templating to generate output project stubs. CI/CD via GitHub Actions.

System Design Studio — <https://www.systemdesign.balashan.dev>

Open-source system-design back-of-envelope and cost-estimation tool with a live formula-driven calculator, Mermaid diagrams, and a React Flow whiteboard. Built with Next.js and Zustand; PostHog for analytics.

ZeroMem — github.com/balashan1/zeromem

Open-source (MIT) agent-memory orchestrator: memory content lives entirely in the user's own Postgres database, with only hashes kept for audit. Ships an MCP server wrapper for direct integration with AI agents.

ChainGuard — github.com/balashan1/chainguard

Decentralized supply-chain insurance and logistics optimizer driven by AI-based anomaly detection for auditors using blockchain and a Tambo-based UI. (static data working model as logistics aggregator's subscriptions for geo APIs are costlier)

EDUCATION

Bachelor of Technology, Electrical & Electronics — Pondicherry University (Jun 2000 – May 2004)